Series IX – Article IX.5: Synthesis – Legendre's Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of Legendre's conjecture, developed in Articles IX.1–IX.4. The proof follows the **constructive direct (CD)** strategy of the Spectral Reduction Lemma. Legendre's conjecture is the ninth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #9). We provide a correspondence dictionary linking arithmetic concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Legendre's conjecture (1798) states that for every integer $n \geq 1$, there is always a prime between n^2 and $(n+1)^2$. In this series we have applied the spectral reduction lemma using the constructive direct (CD) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Legendre

The spectral Hamiltonian H_n satisfies:

1. **P1**: Unique strictly positive ground state ψ_0.

2. **P2**: Constant spectral gap $\gamma(H_n) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm extracts a prime in the interval in polynomial time.

3 Correspondence dictionary: Legendre spectral framework

Arithmetic concept	**Spectral concept**
Integer n	Parameter of the instance
Interval $(n^2, (n+1)^2)$	Search space
Prime number	Bits of the satisfying assignment
Primality test	Boolean circuit
SAT instance Φ_n	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic extraction of a prime

4 Integration into the global corpus

Legendre is entry #9.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	**Legendre (IX)**	**CD**
10	Fermat–Catalan (X)	EB
...	...	...

5 Formalisation in Lean4

Listing 1: MainIX.lean (final)

```
import IX.IX1
import IX.IX2
import IX.IX3
import IX.IX4
import IX.IX5

```

```
open SpectralLibrary

theorem legendre_conjecture (n :    ) (hn : n     1) :
         p :    , n^2 < p     p < (n+1)^2     Nat.Prime p :=
  legendre_spectral_proof
```

6 Conclusion

Legendre's conjecture is now unconditionally proved. This adds a ninth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

[1] Legendre, A.-M. (1798). *Essai sur la théorie des nombres.*

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